



PRODUCT STANDARD
TURBINES AND COMPRESSORS
HYDERABAD

TC-6-2666

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JOB SPECIFICATION OF DRY GAS SEAL

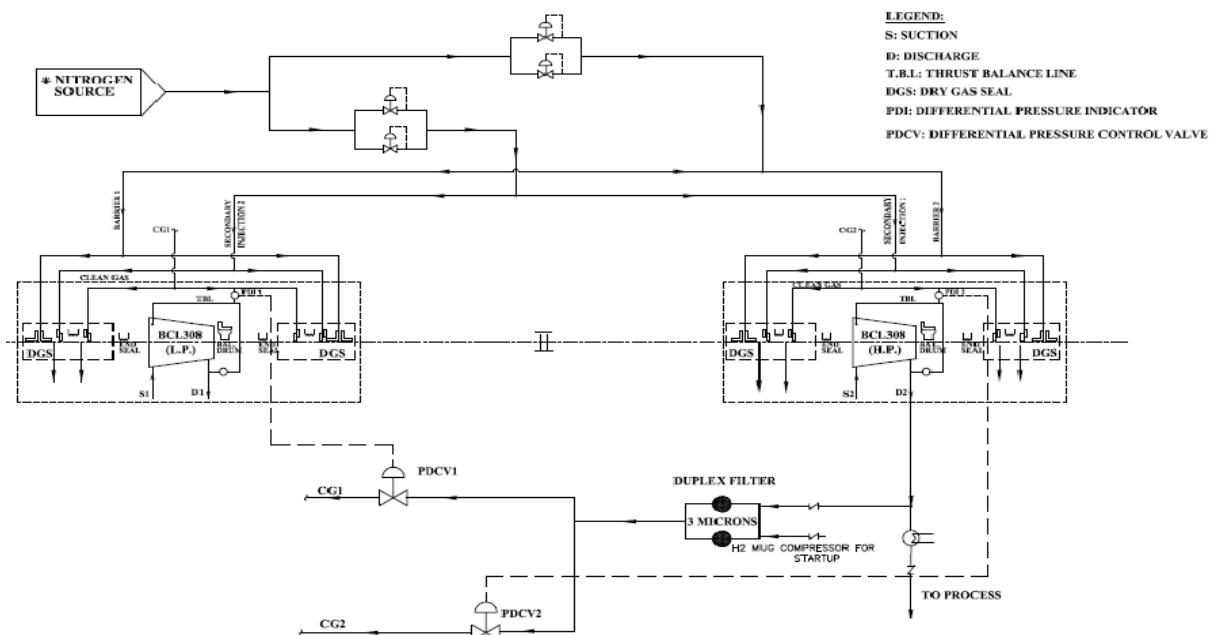
Project: M/s BPCL MUMBAI KHT

1. GENERAL SPECIFICATION:

For general details of dry gas seals, please refer General Specification No.TC52357

For Seal Shaft and Housing Dimensions, please refer Specification No.TC52508

2. SCHEMATIC OF GAS SEAL SYSTEM



Supply of DGS supporting system (Filter / Control skid etc.) is NOT in DGS vendor scope, however DGS vendor shall provide the DGS operating, minimum and maximum operating parameters as indicated in this specification.

3. SEAL PROCUREMENT DETAILS

3.1 DRY GAS SEAL CONFIGURATION

SINGLE [] DOUBLE [] TANDEM [X] TRIPLE []

3.2 DRY GAS SEAL TYPE:

UNIDIRECTIONAL [] BIDIRECTIONAL [X]

3.3 INTERMEDIATE LABYRINTH: Yes / No

3.4 BARRIER SEAL: CARBON RING TERTIARY [X] LABYRINTH SEAL []

Rev. No.	Revisions	Prepared	Approved	Date
00	Refer to record of revisions	RAGHU	YVRL	06.12.2019



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3.5 SEAL CODE DESCRIPTION:

Var	Description	BHEL Material code
01	Tandem type Bi directional Dry Gas Seal ($\Phi 120$) with Barrier seal	TC9762666011
02	Tandem type Bi directional Dry Gas Seal ($\Phi 112$) with Barrier seal	TC9762666020
03	DGS Assy/Dis Assy. Fixture ($\Phi 120$)	TC9762666038
04	DGS Assy/Dis Assy. Fixture ($\Phi 112$)	TC9762666046

4. SEAL HOUSING & SHAFT DIMENSIONS

Please refer TC52508 Var No.18 for DGS DIA 120 (BCL300/A & BCL400 SERIES FOR END MOUNTED BEARING DESIGN) and TC52508 Var No.14 for DGS DIA 112 (BCL300/A & BCL400 SERIES FOR END MOUNTED BEARING DESIGN) for Seal housing and shaft dimensions.

All the interface assembly parts including outer closing cover with its pin, "O" rings, tolerance strips (as applicable), keys, locating/ anti-rotation pins, circlips etc. are in vendor scope.

The outer closing cover shall be provided with the milled projections as shown in the sketch to facilitate free oil drain.

The dimensions indicated in the sketch shall be strictly adhered to.

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5. SEAL DESIGN INPUTS									
5.1. DATA FOR SEAL DESIGN									
		OPERATING CONDITION	NORMAL SOR	RATED	EOR	S/U H2	S/U N2	SOR DOL	EOR DOL
		PARAMETER DESCRIPTION							
		Gas Composition	As per annexure-1						
		Mol. wt.	3.050	3.250	3.250	2.010	28.010	3.050	3.250
		Pr. in the Average Seal gas line upstream of filter (kg/cm ² -a)	63.33	66.73	66.73	52.83	8.50	63.33	66.69
		Temp in Seal gas Line upstream of filter (°C)	123.6	135.1	131.4	96	149	135.9	138.1
		Clean gas pressure (downstream of DPCV1) (kg/cm ² -a)	43.58	42.46	43.58	43.58	4.75	40.61	41.85
		Clean gas temperature (downstream of DPCV1) (°C)	123.6	135.1	131.4	96	149	135.9	138.1
		Clean gas pressure (downstream of DPCV2) (kg/cm ² -a)	53.38	54.24	54.93	48.50	6.58	51.76	53.89
		Clean gas temperature (downstream of DPCV2) (°C)	123.6	135.1	131.4	96	149	135.9	138.1
	Total clean gas leaking into Compr. through end labyrinth seals (both ends) Downstream of DPCV1 (into LP Compressor)	NOR (Kg/hr)	105	105	107	88	99	100	104
		NOR (Am ³ /hr)	27	26	26	31	27	28	27
		Max (Kg/hr)	157	158	161	132	149	141	156
		Max (Am ³ /hr)	40	40	39	47	40	42	40
	Total clean gas leaking into Compr. through end labyrinth seals (both ends) Downstream of DPCV2 (into HP Compressor)	NOR (Kg/hr)	107	110	111	86	109	104	109
		NOR (Am ³ /hr)	22	22	21	28	21	23	22
		Max (Kg/hr)	161	165	167	129	164	156	164
		Max (Am ³ /hr)	33	32	32	41	32	34	33
		Shaft Speed(rpm)	12900	14085	13469	11850	6268	14085	14085
		System settling out press.(kg/cm ² -g)	51.4						
		Rated Speed (rpm)	14085						
		Min. mechanical design pressure for DGS	74						
		Mechanical design temperature for DGS	Min → 0 °C Max → 180 °C						
		Back Pressure in Primary vent	Nor → 0.1kg/cm ² -g / Max → 1.4kg/cm ² -g						

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OPERATING CONDITION		S/U H2 DOL	WITH MUG COMPRESSOR FOR S/U	
PARAMETER DESCRIPTION				
Gas Composition		As per annexure-1	As per annexure-2	
Mol. wt.		2.010	2.09	
Pr. in the Average Seal gas line upstream of filter (kg/cm ² -a)		52.83	66.23	
Temp in Seal gas Line upstream of filter (°C)		111.1	106.0	
Clean gas pressure (downstream of DPCV1) (kg/cm ² -a)		39.65	42.46	
Clean gas temperature (downstream of DPCV1) (°C)		111.1	106.0	
Clean gas pressure (downstream of DPCV2) (kg/cm ² -a)		46.38	53.75	
Clean gas temperature (downstream of DPCV2) (°C)		111.1	106.0	
Total clean gas leaking into Compr. through end labyrinth seals (both ends) Downstream of DPCV1 (into LP Compressor)	NOR (Kg/hr)	82	12	
	NOR (Am ³ /hr)	34	4	
	Max (Kg/hr)	124	19	
	Max (Am ³ /hr)	51	7	
Total clean gas leaking into Compr. through end labyrinth seals (both ends) Downstream of DPCV1 (into LP Compressor)	NOR (Kg/hr)	83	13	
	NOR (Am ³ /hr)	29	4	
	Max (Kg/hr)	124	19	
	Max (Am ³ /hr)	43	5	
Shaft Speed(rpm)		14085	14085	
System settling out press. (kg/cm ² -g)		51.4		
Rated Speed (rpm)		14085		
Min. mechanical design pressure for DGS		74 kg/cm ² -g		
Mechanical design temperature for DGS		Min → 0 °C Max → 180 °C		
Back Pressure in Primary vent		Nor → 0.1kg/cm ² -g / Max → 1.4kg/cm ² -g		



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- Performance Guarantee case is **RATED**
- Rated Speed is 14085 RPM
- Maximum Continuous Speed(MCS) is 14790 RPM
- Minimum Speed is 6268 RPM
- Trip Speed is 14790 RPM
- Slow roll requirement: NA

5.2. AMBIENT CONDITIONS

Temperature	:	Min → 12 °C Nor 35°C Max → 42 °C
Relative Humidity	:	Min. 100 % @ 32°C % Nor. % Max. %

5.3. GAS COMPOSITION FOR VARIOUS CONTINUOUS OPERATING CONDITIONS:

1. Process gas composition: Annexure-1
2. Gas composition for MUG compressor: Annexure -2

5.4. SEALING GAS

Primary Seal	For Normal operation	Dry and filtered process gas			
	For Startup/ emergency	Seal gas from Makeup gas compressor (MUG).			
Secondary Seal		Nitrogen / Plugged			
Barrier Seal		Nitrogen / Instrument Air			
Gas	Parameter	Min	Nor	Max	Mech. Design
Nitrogen	Pressure Kg/Cm² g	3.5	4.5	5.5	10.5
	Temperature oC	-	Amb	-	-
Sweet Gas	Pressure Kg/Cm2 g	NA			
	Temperature oC				
Instrument Air	Pressure Kg/Cm2 g	NA			
	Temperature oC				
MUG Compressor	Disch.Pressure Kg/Cm2 g	64.73			
	Disch.Temperature oC	106			
	The Max.limit on lubricating oil carry over in seal gas stream is 200ppwm				



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6. Applicable Standards / Codes

- 6.1. BHEL General Specification No. TC52357
- 6.2. BHEL Housing Dimension Specification No. TC52508, Var.18 & Var. 14
- 6.3. API 617 – 7th edition
- 6.4. API 614 – 5th edition
- 6.5. NACE MR 0103

7. Special Notes

1. Dry gas seal vendor shall submit written confirmation in their letterhead for the following:
 - a. The spare seal (not installed in casing) shall be suitable for use without any refurbishment for at least 5 years from the date of supply. Vendor shall provide necessary storage conditions to comply with the above.
 - b. If the spare seals are not used at all before 5 years and if the vendor's design standard practice is calling for any refurbishment / revalidation / requalification, after 5 years but before 10 years period vendor shall accept to refurbish the spare seals free of charge for one time to the end customer.
 - c. If the spare seals are used before 5 years, the above Part-b is not applicable.
2. MRT for Dry Gas seals is not applicable as per cl.53 of TC52357. However, supplier to confirm to provide assistance for assembly/dis assembly– 2 visits for a period of 3 days each for each casing at shop.
3. The process gas will contain 4 vol % H₂S for about 24 hours every 3.5 years during sulphiding of the catalyst. Vendor shall confirm that DGS is suitable for this condition.
4. The buffer gas required during start-up/commissioning of Recycle Gas Compressor shall be make up hydrogen from make-up gas compressor (which is a lubricated compressor) at operating pressure of 63.73 kg/cm²A and operating temperature=106 degC. For composition of start-up seal gas refer "MUG process data" enclosed with MR. The maximum limit on lubricating oil carryover in seal gas stream is 200 ppmw. All necessary devices for admitting the clean oil free gas to dry gas seals shall be provided by vendor.

8. DGS support system information requirement:

1. DGS vendor shall provide the following parameters of the DGS (during detail engineering) for LP and HP casing as per table-1.

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TABLE-1																																																																																																																																																						
<table border="1"> <thead> <tr> <th>SI No</th> <th>Parameter</th> <th>HP Casing</th> <th>LP Casing</th> <th>Unit</th> </tr> </thead> <tbody> <tr><td>1</td><td rowspan="3">Primary Seal Gas injection Pressure</td><td>1. Normal</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>2</td><td>2. Low Alarm</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>3</td><td>3. High Alarm</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>4</td><td rowspan="4">Primary Seal Gas Vent Pressure-DE</td><td>1. Normal</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>5</td><td>2. Low Alarm</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>6</td><td>3. High Alarm</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>7</td><td>4. High Trip</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>8</td><td rowspan="4">Primary Seal Gas Vent Pressure-NDE</td><td>1. Normal</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>9</td><td>2. Low Alarm</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>10</td><td>3. High Alarm</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>11</td><td>4. High Trip</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>12</td><td rowspan="3">Primary Seal Gas Vent Flow-DE</td><td>1. Normal</td><td></td><td>Nm3/Hr</td></tr> <tr><td>13</td><td>2. Low Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>14</td><td>3. High Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>15</td><td rowspan="3">Primary Seal Gas Vent Flow-NDE</td><td>1. Normal</td><td></td><td>Nm3/Hr</td></tr> <tr><td>16</td><td>2. Low Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>17</td><td>3. High Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>18</td><td rowspan="3">Secondary Seal Gas Inlet Pressure</td><td>1. Normal</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>19</td><td>2. Low Alarm</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>20</td><td>3. High Alarm</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>21</td><td rowspan="3">Secondary Seal Gas Inlet Flow-DE</td><td>1. Normal</td><td></td><td>Nm3/Hr</td></tr> <tr><td>22</td><td>2. Low Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>23</td><td>3. High Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>24</td><td rowspan="3">Secondary Seal Gas Inlet Flow-NDE</td><td>1. Normal</td><td></td><td>Nm3/Hr</td></tr> <tr><td>25</td><td>2. Low Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>26</td><td>3. High Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>27</td><td rowspan="2">Barriers Seal Gas Inlet Pressure</td><td>1. Normal</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>28</td><td>2. Low Alarm</td><td></td><td>Kg/Cm2g</td></tr> <tr><td>29</td><td rowspan="3">Barriers Seal Gas Inlet Flow</td><td>1. Normal</td><td></td><td>Nm3/Hr</td></tr> <tr><td>30</td><td>2. Low Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>31</td><td>3. High Alarm</td><td></td><td>Nm3/Hr</td></tr> <tr><td>32</td><td>Primary Seal Gas Vent Flow meter</td><td>1. Range</td><td></td><td>Nm3/Hr</td></tr> <tr><td>33</td><td>Secondary Seal Gas Inlet Flow meter</td><td>1. Range</td><td></td><td>Nm3/Hr</td></tr> </tbody> </table>	SI No	Parameter	HP Casing	LP Casing	Unit	1	Primary Seal Gas injection Pressure	1. Normal		Kg/Cm2g	2	2. Low Alarm		Kg/Cm2g	3	3. High Alarm		Kg/Cm2g	4	Primary Seal Gas Vent Pressure-DE	1. Normal		Kg/Cm2g	5	2. Low Alarm		Kg/Cm2g	6	3. High Alarm		Kg/Cm2g	7	4. High Trip		Kg/Cm2g	8	Primary Seal Gas Vent Pressure-NDE	1. Normal		Kg/Cm2g	9	2. Low Alarm		Kg/Cm2g	10	3. High Alarm		Kg/Cm2g	11	4. High Trip		Kg/Cm2g	12	Primary Seal Gas Vent Flow-DE	1. Normal		Nm3/Hr	13	2. Low Alarm		Nm3/Hr	14	3. High Alarm		Nm3/Hr	15	Primary Seal Gas Vent Flow-NDE	1. Normal		Nm3/Hr	16	2. Low Alarm		Nm3/Hr	17	3. High Alarm		Nm3/Hr	18	Secondary Seal Gas Inlet Pressure	1. Normal		Kg/Cm2g	19	2. Low Alarm		Kg/Cm2g	20	3. High Alarm		Kg/Cm2g	21	Secondary Seal Gas Inlet Flow-DE	1. Normal		Nm3/Hr	22	2. Low Alarm		Nm3/Hr	23	3. High Alarm		Nm3/Hr	24	Secondary Seal Gas Inlet Flow-NDE	1. Normal		Nm3/Hr	25	2. Low Alarm		Nm3/Hr	26	3. High Alarm		Nm3/Hr	27	Barriers Seal Gas Inlet Pressure	1. Normal		Kg/Cm2g	28	2. Low Alarm		Kg/Cm2g	29	Barriers Seal Gas Inlet Flow	1. Normal		Nm3/Hr	30	2. Low Alarm		Nm3/Hr	31	3. High Alarm		Nm3/Hr	32	Primary Seal Gas Vent Flow meter	1. Range		Nm3/Hr	33	Secondary Seal Gas Inlet Flow meter	1. Range		Nm3/Hr	
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ANNEXURE - 1
GAS COMPOSITION

OPERATING CONDITIONS (Continued) (1-2.1.1.1) (1-3.1.2) (1-3.1.3)							
GAS ANALYSIS:		Normal SOR (P2)	Rated Rated (P4)	Other Conditions			REMARKS:
● MOL % ○				EOR (P3)	S/U H2 (P5)	S/U N2 (P6)	
	MW						
AIR	28.97	0,00	0,00	0,00			
OXYGEN	32,00	0,00	0,00	0,00			
NITROGEN	28,02	0,04	0,03	0,03		>99,99	
WATER VAPOR	18,02	0,58	0,59	0,59			
CARBON MONOXIDE	28,01	0,01	0,00	0,00			
CARBON DIOXIDE	44,01	0,00	0,00	0,00			
HYDROGEN SULFIDE	34,08	0,012	0,016	0,016			(P9) (1-2.2.1.6)
HYDROGEN	2,02	94,95	94,12	94,12	>99,50		(1-2.2.1.9)
METHANE	16,04	3,37	3,83	3,83			
ETHYLENE	28,00	0,00	0,00	0,00			
ETHANE	30,07	0,55	0,79	0,79			
PROPYLENE	42,08	0,00	0,00	0,00			
PROPANE	44,09	0,20	0,28	0,28			
i-BUTANE	58,12	0,03	0,04	0,04			
n-BUTANE	58,12	0,05	0,07	0,07			
i-PENTANE	72,15	0,00	0,00	0,00			
n-PENTANE	72,15	0,00	0,00	0,00			
HEXANE PLUS		0,21	0,22	0,22			
AMMONIA	17,03	0,00	0,00	0,00			
CORROSIVE AGENTS		0,00	0,00	0,00			(1-2.2.1.3)
TOTAL		100	100	100			
AVG. MOL. WT.		3,05	3,25	3,25	2,09	28	

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Annexure-2

MUG COMPRESSOR GAS COMPOSITION

1	GAS ANALYSIS AT OPERATING CONDITIONS				
2	MOLE % (BY VOLUME) ONLY				
3	● SERVICE / ITEM NO.	Make UP Gas Compressor (Continuous)			
4	● STAGE	1st Stage	1st Stage	1st Stage	
5	● NORMAL OR ALT.	SOR (P2)	Rated (P4)	EOR (P3)	S/U N2 (P11)
6	M.W.				
7	AIR 28.966	0.00	0.00	0.00	
8	OXYGEN 32	0.00	0.00	0.00	
9	NITROGEN 28.016	0.005	0.005	0.005	>99.99
10	WATER VAPOUR (Note-P12) 18.016	0.005	0.005	0.005	
11	CARBON MONOXIDE (Note-P8) 28.01	0.00	0.00	0.00	
12	CARBON DIOXIDE (Note-P8) 44.01	0.00	0.00	0.00	
13	HYDROGEN SULFIDE 34.076	0.00	0.00	0.00	
14	HYDROGEN 2.016	99.5	99.5	99.5	
15	METHANE 16.042	0.49	0.49	0.49	
16	ETHYLENE 28.052	0.00	0.00	0.00	
17	ETHANE 30.068	0.00	0.00	0.00	
18	PROPYLENE 42.078	0.00	0.00	0.00	
19	PROPANE 44.094	0.00	0.00	0.00	
20	I-BUTANE 58.12	0.00	0.00	0.00	
21	n-BUTANE 58.12	0.00	0.00	0.00	
22	I-PENTANE 72.146	0.00	0.00	0.00	
23	n-PENTANE 72.146	0.00	0.00	0.00	
24	HEXANE PLUS	0.00	0.00	0.00	
25	AMMONIA 17.031	0.00	0.00	0.00	
26	HYDRO. CHLORIDE 36.461	0.00	0.00	0.00	
27	CHLORINE 70.914	0.00	0.00	0.00	
28	CHLORIDES - TRACES (Note-P9)	0.00	0.00	0.00	
29	OTHERS				
30	TOTAL	100	100	100	
31					
32	■ CALCULATED MOL WT.	2.09	2.09	2.09	28.00
33	■ Cp/Cv (K) °C				



PRODUCT STANDARD
TURBINES AND COMPRESSORS
HYDERABAD

TC-6-2666

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RECORD OF REVISIONS

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